BioHacks

McMaster University, Hamilton, Ontario

MLH MAJOR LEAGUE HACKING

BIOHACKS 2026

Hacker Package

November 7-8, 2026

SHIONOGI

ALS SLA CANADA

BMETT

WELCOME TO BIOHACKS 2026

You made it. For the next two days you and your team will take on one patient case, choose a track and build something for him.

This package holds the schedule, the prizes, how to submit and the full case file. Read the case first. Then go build.

Discord

Every announcement this weekend goes out on the BioHacks Discord.
discord.com/invite/mGZRP7vpj

Instagram

Follow

@_biohacks

and tag us in your posts from the weekend.

Organizers

Find us at the check-in desk on the 2nd floor, outside the MCCD 2100 auditorium, or email bioeng4youth@gmail.com.

SAFETY AND HELP

McMaster Security, 24 hours

905-522-4135

From any campus phone, dial **88**

EFRT, McMaster's Emergency First Response Team, is a team of student first responders on campus. Reach them through Security at 905-522-4135 or 88.

Emergency 911

After you call, tell the nearest organizer.

Code of conduct

BioHacks follows the MLH Code of Conduct. Harassment of any kind is not tolerated. Report a concern to any organizer, or to MLH at +1 343 453 4532 or incidents@mlh.io.

Fire alarm

If the alarm sounds, leave by the nearest exit and wait outside until an organizer says it is safe to return.

Overnight

Hacking runs through the night. The MCCD 2100 auditorium is open for sleeping. Keep your laptop and valuables with you or with a teammate.

Halal and vegetarian options are served at every meal.

SATURDAY, NOVEMBER 7

Ceremonies · Guest lectures · Workshops · Food · Judging · Check-in, booths and deadline

Schedule

8 AM

9 AM
Doors open and check-in
2nd floor, outside the MCCD 2100 auditorium

10 AM
Opening ceremony
MCCD 2100 auditorium

11 AM
Team formation — **Team registration closes at 12:00 PM**
Anyone without a team is matched with one here.

12 PM
Lunch
Arts Quad

1 PM
CAD for Medical Devices, from Sketch to 3D Print
Workshop, MCCD 3200

2 PM
Hacking with GitHub Copilot — MLH workshop, MCCD 7150

Intro to Google AI Studio — MLH workshop, MCCD 7150

3 PM
Guest lecture with Dr. Tandon
GUS Medical and Veterinary Schools, MCCD 3200

4 PM

5 PM
Guest lecture with Dr. Rock
UHN Michener Institute of Education, MCCD 7140

6 PM
Pitching workshop
MCCD 2100 auditorium

7 PM

8 PM
Dinner — Arts Quad

TechTogether Meetup
MLH, MCCD 7140

9 PM

10 PM

11 PM

12 AM
Midnight snack — MCCD 2100 auditorium

Team form

Sponsor booths
11:00 AM to 6:30 PM
2nd floor, outside the auditorium

SUNDAY, NOVEMBER 8

Ceremonies | Guest lectures | Workshops | Food | Judging | Check-in, booths and deadline

Schedule

8 AM

Breakfast
MCCD 2100 auditorium

9 AM

10 AM

11 AM

Submissions due on Devpost at 11:30 AM

12 PM

Lunch
Arts Quad

1 PM

Expo judging
MCCD 7110 and 7120

2 PM

3 PM

4 PM

Closing ceremony
MCCD 2100 auditorium

5 PM

Event ends

6 PM

SUBMISSION AND JUDGING

PROJECTS ARE DUE ON DEVPOST BY 11:30 AM ON SUNDAY

1. **Team up.** Anyone without a team is matched with one during team formation at 11:00 AM on Saturday. Form a team of 2 to 4 and lock in one track on the team form by 12:00 PM on Saturday. Each team competes in exactly that one track and is judged only in it.

2. **Build.** Workshops, guest lectures, the MLH hardware lab and our 3D printers are open to every team. Ask an organizer to print a part.

3. **Submit.** One project per team on Devpost by 11:30 AM on Sunday at **biohacksmcmaster.devpost.com**.

   External sponsor prizes are optional. A team can opt in to any of them on Devpost and win them alongside its track result.

   Devpost

4. **Present.** Expo judging runs from 1:30 to 3:00 PM on Sunday in MCCD 7110 and 7120. Judges come to your table.

5. **Celebrate.** Winners are announced at the closing ceremony at 4:30 PM in the MCCD 2100 auditorium.

ALEX CHEN

Alex is 47. He has taught senior biology at a high school on the Hamilton mountain for nineteen years, writes everything on the whiteboard and walks the rows while students work. He coached cross-country for a decade and ran half-marathons until the spring of 2025.

He lives in Westdale with his partner Jordan, a medical radiation technologist on 7 am to 4 pm shifts, and their two children, Maya, 16, and Leo, 14. In June 2024 a student asked whether his marker had run dry. His handwriting had started to fade out halfway across the board. The marker was fine. He waited a few months, then saw his family doctor. What follows is his file.

Referral
14 October 2024
Alex Chen, 45, male
To: Plastic and Hand Surgery Clinic

Mountain Brow Family Health Team

Hamilton, Ontario

Re: Right hand weakness and deteriorating handwriting, four months

Thank you for seeing Alex, a 45-year-old right-handed teacher who is otherwise well. Since June he has noticed that his handwriting fades across the page and that his right thumb and index finger tire quickly when writing on a whiteboard. He has started dropping his whiteboard marker and his coffee cup. No pain. No numbness or tingling. He reports some cramping in the right forearm at night.

On examination the right thenar eminence looks a little flatter than the left. Grip is slightly weaker on the right. Phalen's test was equivocal. Sensation to light touch was normal in all fingers. He works at a desk in the evenings marking, and I suspect a carpal tunnel syndrome or a median neuropathy at the wrist.

I would be grateful for your assessment and for nerve conduction studies at the wrist. He has no significant past history, takes no medication, does not smoke and runs regularly.

Dr. Priya Raman, MD, CCFP
Family physician

Consultation 9 December 2024
Follow-up 21 April 2025
Alex Chen
Copy to: Dr. P. Raman

Plastic and Hand Surgery Clinic

Consultation and follow-up. Dictated, not read.

Re: Query right carpal tunnel syndrome

Consultation. Six months of right hand weakness affecting fine tasks, no sensory symptoms. Nerve conduction across the right wrist: median sensory and motor latencies within normal limits; no slowing across the carpal tunnel. Visible wasting of the right thenar muscles and the muscle between thumb and index finger. Grip 4 out of 5 on the right. Given the thenar wasting, his occupational demands and his wish to try something, I offered a right carpal tunnel release as a reasonable trial of treatment. He elected to proceed after discussion of the risks and the uncertain benefit. Surgery 20 January 2025, uncomplicated.

Follow-up, three months after surgery. Wound well healed. He reports no improvement. If anything the hand is weaker, and he now notices twitching under the skin of the forearm, which he describes as "rippling". He has also begun to mention his right foot catching on stairs. I have no further surgical options to offer. Returned to the care of his family physician with a suggestion that a neurological opinion may be helpful if symptoms progress.

Dr. Marcus Oyelaran, MD, FRCSC
Plastic and reconstructive surgery

10 February 2026
Alex Chen
12 August 2025 to 5 February 2026, 18 sessions
To: Dr. P. Raman

Westdale Physiotherapy and Sports Clinic

Discharge summary

Re: Right foot drop, treated as peroneal weakness. Not improving.

Alex was referred in August 2025 for a right foot that catches on curbs and slaps on landing. He had stopped running in May because the right leg "felt heavy and would not do what I asked". A lumbar spine MRI in December 2025 was reported as unremarkable for his age, with no nerve root compression.

Over six months of strengthening, balance work and a trial of an off-the-shelf ankle brace, his right ankle dorsiflexion has gone from 4 out of 5 to 3 out of 5. Calf and thigh circumference on the right are each about 2 cm smaller than the left. He now uses a cane outside the house. He reports two falls, both in the evening, one on the stairs at home and one leaving the school at dusk. He has stopped cycling to work and drives instead.

Three things concern me. The weakness is progressing despite treatment. His reflexes are brisk, including at the right ankle, which is not what I would expect from a peripheral nerve problem. And he mentions that his right hand has been weak for well over a year and that a wrist operation did not help it. I do not think this is an ankle problem. I would strongly recommend a neurology referral, and I have told Alex so.

Daniela Ferreira, PT, MScPT
Registered physiotherapist

Referral, marked urgent
16 February 2026
Alex Chen, 46, male
To: Neurology, outpatient clinic

Mountain Brow Family Health Team

Hamilton, Ontario

Re: Progressive painless weakness, right arm and leg, twenty months

I would be grateful if you could see Alex soon. I first saw him in October 2024 with right hand weakness that I referred as a carpal tunnel syndrome. His wrist studies were normal, a carpal tunnel release in January 2025 did not help, and in the past year he has developed a right foot drop that physiotherapy has not improved and that a lumbar MRI does not explain.

Today he reports muscle cramps, visible twitching in both forearms and lately the right thigh, and that he has lost about 6 kg without trying since the summer. His voice is unchanged as far as he is aware, although his partner says he is quieter in the evenings. There is no pain, no numbness, no change in bladder or bowel function and no visual symptoms. On examination he has wasting of the right hand and brisk reflexes in all four limbs. I am concerned about a progressive neurological condition and would value your opinion. I am sorry this has taken as long as it has.

Dr. Priya Raman, MD, CCFP

Seen 11 June 2026
Referral received 18 February 2026
Alex Chen, 47, male
Referring: Dr. P. Raman

Neurology Outpatient Clinic

Consultation note. Neuromuscular service.

Reason for referral: progressive painless weakness, right arm and leg, now two years

History

Right-handed, 47. First symptom June 2024: handwriting fading on the whiteboard, fatigue in the right thumb and index finger. Right carpal tunnel release January 2025 after normal wrist studies, no benefit. Right foot drop from spring 2025, progressing through physiotherapy; lumbar MRI unremarkable. Cramps in both calves and the right forearm, worse at night. Fasciculations noticed in the forearms since April 2025, now also in the right thigh. Unintentional weight loss, 86 kg two years ago to 78 kg today.

On direct questioning: speech is clear in the morning but his partner and students notice slurring by late afternoon, which he had put down to tiredness. Occasional coughing when drinking water quickly. He has started sleeping on two pillows because he feels "short of air" lying flat. No diplopia, no sensory symptoms, no sphincter change. Mood is low but he denies hopelessness. He is still teaching full time and still drives.

Family history: no neurological disease in parents, siblings or grandparents. No dementia. Social: lives with partner and two teenagers, non-smoker, two drinks a week. No toxin exposure.

Examination

General. Thin. Weight 78 kg. Normal cognition in conversation, but on a bedside screen he produced only nine words beginning with the same letter in one minute, with intact memory and orientation.

Cranial nerves. Eye movements full. Face symmetric. Tongue shows mild wasting with fasciculations at rest. Jaw jerk brisk. Palate elevates symmetrically. Speech mildly spastic in quality late in the consultation, intelligible.

Upper limbs. Marked wasting of the right thenar eminence and first dorsal interosseous, with relative sparing of the hypothenar muscles. Fasciculations in both forearms and the right deltoid. Tone increased on the right. Power: right grip 4/5, right finger abduction 3/5, right thumb abduction 3/5, left upper limb 5/5. Reflexes brisk and symmetric, right more than left. Hoffmann sign present on the right.

Lower limbs. Right calf wasting. Fasciculations in the right quadriceps. Tone increased bilaterally. Power: right ankle dorsiflexion 3/5, right hip flexion 4/5, left lower limb 5/5. Reflexes brisk, with three beats of clonus at the right ankle. Right plantar response extensor, left flexor.

Sensation. Normal to light touch, pinprick, vibration and proprioception throughout. **Gait.** Right foot drop with a high-stepping gait and a cane. Unable to walk on heels on the right. Romberg negative.

Investigations requested

Nerve conduction studies and needle electromyography, four regions. MRI brain and cervical spine. Bloods as per neuromuscular panel. Pulmonary function tests upright and supine. Referral to speech-language pathology. Review in four weeks with results.

Dr. Hannah Weiss-Okafor, MD, FRCPC
Neurologist, neuromuscular disease

Electromyography Laboratory

Nerve conduction studies and needle EMG

18 June 2026
Alex Chen
Requested by: Dr. H. Weiss-Okafor

Nerve conduction studies

Nerve	Result	Reference
Right median motor, to thumb muscles	Amplitude 1.6 mV, latency and velocity normal	Amplitude above 4.0 mV
Right ulnar motor, to little-finger muscles	Amplitude 5.5 mV, velocity normal, no conduction block	Amplitude above 6.0 mV
Right peroneal motor, to foot	Amplitude 0.9 mV, velocity normal	Amplitude above 2.0 mV
Left median and ulnar motor	Amplitudes at lower limit of normal	
Sensory studies, median, ulnar, sural, both sides	All normal	Normal

Needle electromyography

Region	Muscles sampled	Findings
Bulbar	Tongue	Fasciculation potentials, fibrillations, enlarged motor units
Cervical	Right first dorsal interosseous, right biceps, left deltoid	Fibrillations and positive sharp waves, fasciculations, large polyphasic motor units with reduced recruitment, right more than left
Thoracic	Right T9 paraspinal	Fibrillations, fasciculations
Lumbosacral	Right tibialis anterior, right vastus lateralis, left gastrocnemius	Active denervation and chronic reinnervation in all three; severely reduced recruitment in tibialis anterior

Summary

Widespread active denervation and chronic reinnervation in four body regions, including clinically unaffected muscles, with normal sensory conduction and no conduction block. There is no electrophysiological evidence of a peripheral neuropathy, a disorder of the neuromuscular junction, or a myopathy.

Dr. Hannah Weiss-Okafor, MD, FRCPC

Laboratory Results

Neuromuscular panel, venous blood

Collected 12 June 2026
Alex Chen

Test	Result	Reference
Creatine kinase	470 U/L	Below 195 U/L
Complete blood count, electrolytes, renal and liver function, glucose, calcium, thyroid, vitamin B12	All within reference range	
Serum protein electrophoresis and immunofixation	No paraprotein	None
HIV, syphilis, Lyme serology; anti-GM1 antibodies; heavy metals	Negative	Negative
Serum neurofilament light chain	88 pg/mL	Below 15 pg/mL at age 47
Genetic panel: C9orf72 repeat expansion, SOD1, TARDBP, FUS and 28 further genes	No pathogenic variant detected	Reported 30 July 2026

Pulmonary Function Laboratory

Spirometry, upright and supine

18 June 2026
Alex Chen

Measure	Result	Reference
Forced vital capacity, sitting	78 per cent of predicted	Above 80 per cent
Forced vital capacity, lying flat	64 per cent of predicted	Within 10 per cent of the sitting value
Sniff nasal inspiratory pressure	52 cm H_2O	Above 70 cm H_2O

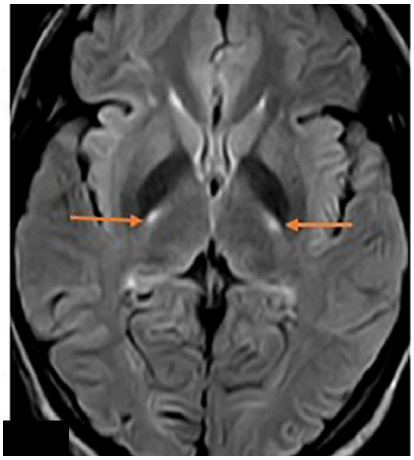
Comment: the fall in vital capacity on lying flat indicates diaphragm weakness.

Diagnostic Imaging

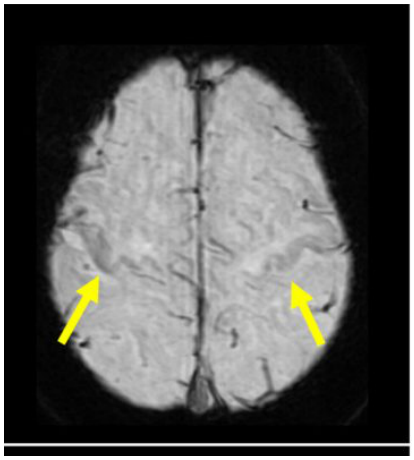
MRI brain and cervical spine, with and without contrast

16 June 2026
Alex Chen
Requested by: Dr. H. Weiss-Okafor

Clinical information: progressive upper and lower limb weakness with brisk reflexes. Exclude cord compression. Query motor neuron disease.



A. Brain, axial FLAIR at the level of the internal capsule. Symmetric hyperintensity along the posterior limb of the internal capsule on both sides (arrows), following the corticospinal tracts.



B. Brain, axial susceptibility-weighted near the vertex. A thin band of low signal along the precentral gyrus on both sides (arrows), sometimes called the motor band sign.

C. Cervical spine, sagittal T2. Normal alignment. No disc protrusion, canal stenosis, cord compression or intrinsic cord signal change from the foramen magnum to T2.

Impression

- Bilateral, symmetric corticospinal tract signal change and a motor band sign. These findings are supportive of a degenerative process affecting the upper motor neurons in the appropriate clinical context. They are not specific.
- No structural lesion of the brain or cervical cord to account for the patient's weakness. Specifically, no cervical spondylotic myelopathy, no demyelinating lesion, no mass.
- No abnormal enhancement after contrast, so nothing to suggest tumour, infection or inflammation.

Dr. Thanh Nguyen, MD, FRCPC
Neuroradiology

Neurology Outpatient Clinic

Letter to referring physician. Copy given to the patient at his request.

14 July 2026
Alex Chen, 47, male
To: Dr. P. Raman

Diagnosis: amyotrophic lateral sclerosis (ALS), limb onset, with early bulbar and respiratory involvement

Dear Dr. Raman,

I saw Alex and his partner Jordan today to go through his results, and I am sorry to say that the diagnosis is ALS. I spent the appointment with them and the clinic social worker, and they will return in two weeks to talk further. He asked that this letter be written in plain language and that he be given a copy, which I have done.

How the diagnosis was made. ALS is diagnosed on the clinical picture, not on any single test. Alex has progressive weakness with signs of both upper motor neuron damage (brisk reflexes, clonus, a Hoffmann sign, an upgoing plantar, spastic speech) and lower motor neuron damage (wasting, fasciculations, low motor amplitudes, denervation on EMG) in the same regions, now spread across all four: bulbar, cervical, thoracic and lumbosacral. Sensation is normal. The MRI excludes the structural conditions that can mimic this, the bloods exclude the treatable ones, and the EMG shows the process is active and widespread. He meets the Gold Coast criteria without ambiguity. In retrospect he probably met them a year ago.

What ALS is. A progressive loss of the motor neurons that drive voluntary muscle, in the brain and the spinal cord. The muscle itself is healthy; it wastes because the nerve supply dies. Sensation, eye movements, bladder and bowel are spared. Thinking is usually spared, though about half of patients develop some change in word-finding or executive function, and Alex's reduced verbal fluency is of that kind. The cause in his case, as in about nine of ten cases, is unknown.

Treatment. There is no cure. Two drugs have been shown to slow progression modestly: riluzole, which I have started today, and edaravone, which we will discuss. A third drug, tofersen, is the first treatment to act on a cause of ALS. It is an antisense oligonucleotide that silences a faulty SOD1 gene, and because it cannot cross the blood-brain barrier it is given by lumbar puncture every month. Health Canada authorised it with conditions in February 2025. It applies to the roughly two per cent of patients who carry a SOD1 mutation. Alex's genetic panel was negative, so, like most people with this disease, he has no treatment directed at its cause. Beyond drugs, what has been shown to lengthen life is care at a multidisciplinary clinic, keeping his weight up, and breathing support at night once it is needed.

What to expect. His progression over two years has been steady rather than rapid, which is in his favour. The pattern will continue: the right hand and leg will weaken further and the left side will follow; speech, swallowing and breathing have begun to be involved and will progress. I have been honest with them that the median survival from symptom onset is two to four years, that the range is wide, and that the decisions ahead (a feeding tube, ventilation at night, how he wants to communicate when speech is difficult) are better made early. I have referred him to the regional ALS clinic, which will see him every three months, and to ALS Canada for equipment and support.

He intends to keep teaching this year. I see no reason he should not.

Dr. Hannah Weiss-Okafor, MD, FRCPC
Neurologist, neuromuscular disease

Medication	Riluzole 50 mg twice daily since July. Tolerating. Edaravone declined for now.
Living situation	Two-storey house, bedroom and only full bathroom upstairs, fourteen stairs with a rail on the left. Partner works 7 am to 4 pm; Alex is alone at home on his non-teaching days and from the end of school until 4:30 pm. Children are at school until 3 pm.
Work	Full-time teaching, four classes a day. School has agreed to a stool and a microphone on request; he has not asked for either.

Neurology

ALSFRS-R 32 of 48 (bulbar 9, fine motor 6, gross motor 7, breathing 10). Decline of about 0.6 points per month since onset, roughly linear. Right hand now 3/5 grip, cannot sustain a pinch for more than a few seconds. Left hand weaker than in June: grip 4/5, early first dorsal interosseous wasting. Cognitive screen (ECAS) 101 of 136, with verbal fluency and executive scores below cut-off and memory, language and visuospatial scores normal.

Speech-language pathology

Speech is intelligible but effortful by late afternoon. Reading passage: 165 words per minute at 9 am, 118 at 4 pm; sentence intelligibility 98 per cent in the morning, 91 per cent in the afternoon. He reports that students at the back of the room have started asking him to repeat himself after lunch and that he has stopped doing the end-of-day recap. Voice banking discussed; he has not started and is reluctant ("it feels like giving up"). Timed water swallow: cough on thin liquid; solids and thickened liquids unaffected. He has started avoiding eating in the staff room.

Dietitian

Weight 77.5 kg, from 86 kg. Meals take 35 to 40 minutes; he tires before finishing. Right-handed cutlery no longer practical; he has been eating with a spoon in the left hand. Calorie supplement started. Feeding tube discussed as a future decision while breathing capacity is still above 50 per cent. He says Sunday cooking for the family is the thing he least wants to give up.

Occupational therapy

Handwriting now illegible after about ten minutes. Marks by typing with the left hand and two right fingers; phone use is one-handed and slow. Buttons, zips and shoelaces take help in the morning. Drives an automatic with the left foot off the pedals, which he has not told his insurer. Stairs are the main hazard: both falls were after 6 pm on the stairs or at dusk. Advised against carrying anything on the stairs. Home assessment booked. He has declined a stairlift "for now".

Physiotherapy and respiratory

10-metre walk with cane 14.2 seconds. Ankle-foot orthosis fitted on the right, which he wears to school but not at home. Forced vital capacity 76 per cent sitting, 61 per cent supine. Sleeps on two pillows, wakes once or twice with a dry mouth. Non-invasive ventilation discussed; not yet indicated. Cough peak flow 310 L/min.

Social work

Has disclosed to the school and to the children. Finances adequate while he works; partner's shifts cannot easily change. Wants to be "reachable and useful" while alone in the afternoons. Would like to get to his daughter's graduation in two years without a wheelchair.

MEASUREMENTS

Measure	Alex	Typical
Hand		
Grip strength, dynamometer	Right 14 kg, left 32 kg	About 48 kg
Key pinch	Right 3 kg, left 8 kg	About 11 kg
Nine-hole peg test	Right 44 s, left 23 s	About 20 s
Speech and swallowing		
Speaking rate, same passage, 9 am and 4 pm	165 and 118 words per minute	150 to 190
Sentence intelligibility, 9 am and 4 pm	98 and 91 per cent	Above 97 per cent
Timed water swallow, 100 mL	Cough on thin liquid	No cough
Breathing		
Forced vital capacity, sitting and lying flat	76 and 61 per cent of predicted	Above 80; within 10 per cent of sitting
Walking		
10-metre walk with cane	14.2 s	Under 8 s
Falls in the past three months	Two, both after 6 pm	None
Progression		
ALSFRS-R at 0, 6, 12, 18, 24 and 27 months	48, 43, 39, 36, 33, 32	48
Serum neurofilament light chain	88 pg/mL	Below 15 pg/mL at his age
Weight	77.5 kg; 86 kg two years ago	

FROM ALEX

You have just read two years of my life in about twenty minutes.

I am still me. I still get up at six and I still argue about the Leafs. This thing has taken my right hand and some of my voice, and it is going to take more. The people around me are still here. My wife still laughs at my jokes. My kids still climb on me. My students still email me at midnight.

I want to be in a classroom in June. I want to be the one who cooks on Sunday, even if it takes me all afternoon. I want my kids to remember me as their dad and not as a patient.

Whatever you build this weekend, I hope it comes from a place of caring about someone like me. If it helps one person hold on to one of those ordinary things a little longer, you will have done something that matters to a whole family. Thank you for spending your weekend on us.

Alex

THE TRACKS

Choose one track to focus on.

1. Detection and Diagnostics

- Alex waited two years for a diagnosis, through four clinicians and an operation that did nothing.
- In Canada the wait is typically 15 to 27 months depending on the province.
- Since diagnosis his disease is followed with a questionnaire every three months.
- Build something that detects the disease earlier or predicts how it will progress. Use the datasets on the next page or find your own.

2. Therapeutics and Delivery

- The one drug that acts on a cause of ALS applies to about two in a hundred patients. Alex is not one of them.
- The two other approved drugs add months. More than fifty late-stage trials have failed since riluzole was approved in 1995.
- Getting anything into a spinal motor neuron is itself an unsolved problem.
- Propose a therapy, or a way of delivering one, that could slow the disease or protect the neurons that remain. Say what it would mean for Alex now and for someone where he was in 2024.

3. Wearable Devices and Quality of Life

- He is strong in the morning and weak by the afternoon. He teaches in a classroom where he needs to be heard, he is alone at home until 4:30, and he has fallen on the stairs.
- His right hand has a few seconds of pinch left. His left is starting to go.
- Most assistive devices end up in a drawer because nobody asked the person what they wanted.
- Design a device or tool that lets Alex keep one thing that matters to him as his abilities change. The MLH hardware lab is open all weekend if you want to build.

DATA

Dataset	What it is	Access
ALS Focus, Harvard Dataverse	Survey data from the ALS Association: self-reported ALSFRS-R and vital capacity over time (892 records), demographics for 1,545 people with ALS and caregivers, health status and telehealth responses, with a data dictionary.	Open download after accepting the data use terms. No account.
ALS progression data, Efron and Hastie	1,822 people with ALS from the 2012 DREAM challenge, drawn from clinical trial records: demographics, onset site, symptoms, ALSFRS items, vital capacity, vitals and weight summarised per patient, with the rate of ALSFRS decline (dFRS) as the target. 371 columns. Documented in the CASIdata R package.	Open download, one text file. No account.
Gait in Neurodegenerative Disease, PhysioNet	Foot-force and stride-interval recordings from 13 people with ALS, 15 with Parkinson's, 20 with Huntington's and 16 controls.	Open download, no account.
VOC-ALS, Synapse	1,224 recordings of sustained vowels and syllable repetition from 102 people with ALS and 51 matched controls, labelled with ALSFRS-R speech score and dysarthria severity. Paper: Scientific Data 2024.	Free Synapse account (registered user). Synapse terms of use.
bigP3BCI, PhysioNet	P300 speller EEG from 326 participant sessions across 20 studies, including people with ALS with ALSFRS-R in the file headers. 10 GB zipped; single studies can be pulled separately.	Open download. CC BY 4.0.
EEGET-ALS, Figshare	32-channel EEG and eye tracking from six people with ALS using an eye-gaze speller at home over several months, and 170 controls. 1.6 GB. Paper: Scientific Data 2024.	Open download, no account. CC0.
Seattle ALS Patient Profile Database	Interviews every three months for 18 months with 143 people with ALS and 123 partners: coping, support, devices, psychological health, outcomes. Mendeley Data.	Open download. CC BY-NC 3.0.

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A note on the images

The MRI images on page 12 are reproduced from open-access teaching material. They illustrate the findings described in the report and are not from one patient. Alex, his family and every clinician and practice in this file are fictional.

JUDGING RUBRIC

DETECTION AND DIAGNOSTICS

Criterion	Level 4	Level 3	Level 2	Level 1
Clinical grounding **20**	**20 points** • Clear connection to the patient's findings, abilities and priorities • Design choices follow from the case	**15 points** • Good connection to the case • Some design choices not tied to it	**10 points** • Loose connection to the case • Could apply to any patient with this condition	**5 points** • Little or no connection to the case
Diagnostic value **30**	**30 points** • Clear explanation of what is detected and why it matters • Clear advantage over current practice	**22 points** • Good explanation of what is detected • Advantage over current practice stated with minor gaps	**15 points** • Adequate explanation of what is detected • Advantage over current practice unclear	**8 points** • Unclear what is detected or why it matters
Evidence and testing **20**	**20 points** • Strong reasoning for why it works, supported by data, measurements or a prototype • Clear plan for how the method would be tested and accuracy measured	**15 points** • Good reasoning with some supporting evidence • Good testing plan with minor gaps	**10 points** • Reasoning present but little evidence • Testing mentioned but not developed	**5 points** • No clear reasoning or evidence • No testing considered
Feasibility **15**	**15 points** • Clear and realistic path to use, with cost, access and regulation considered	**11 points** • Good consideration of practical barriers with minor gaps	**8 points** • Practical barriers acknowledged but not addressed	**4 points** • Practical barriers missing or ignored
Communication **15**	**15 points** • Clear, organised and concise • Questions answered well	**11 points** • Clear overall with minor issues • Most questions answered	**8 points** • Hard to follow in places • Questions partly answered	**4 points** • Unclear or disorganised • Questions not answered

Judges score each project out of 100 with this rubric, then name the best, second best and third best project they saw in the track. The rankings from all judges determine which two teams win in each track.

JUDGING RUBRIC

THERAPEUTICS AND DELIVERY

Criterion	Level 4	Level 3	Level 2	Level 1
Clinical grounding **20**	**20 points** • Clear connection to the patient's findings, abilities and priorities • Design choices follow from the case	**15 points** • Good connection to the case • Some design choices not tied to it	**10 points** • Loose connection to the case • Could apply to any patient with this condition	**5 points** • Little or no connection to the case
Mechanism and rationale **30**	**30 points** • Clear and well supported explanation of how the treatment acts on the condition • Scientifically sound	**22 points** • Good explanation of the mechanism with minor gaps or light support	**15 points** • Mechanism stated but weakly explained or supported	**8 points** • Mechanism unclear or missing
Reaching the patient **20**	**20 points** • Clear explanation of how the treatment reaches where it acts • Main risks identified and addressed	**15 points** • Good explanation of delivery • Risks identified with minor gaps	**10 points** • Delivery or risks mentioned but not developed	**5 points** • Delivery and risks not considered
Feasibility **15**	**15 points** • Clear and realistic path to use, with cost, access and regulation considered	**11 points** • Good consideration of practical barriers with minor gaps	**8 points** • Practical barriers acknowledged but not addressed	**4 points** • Practical barriers missing or ignored
Communication **15**	**15 points** • Clear, organised and concise • Questions answered well	**11 points** • Clear overall with minor issues • Most questions answered	**8 points** • Hard to follow in places • Questions partly answered	**4 points** • Unclear or disorganised • Questions not answered

Judges score each project out of 100 with this rubric, then name the best, second best and third best project they saw in the track. The rankings from all judges determine which two teams win in each track.

JUDGING RUBRIC

WEARABLE DEVICES AND QUALITY OF LIFE

Criterion	Level 4	Level 3	Level 2	Level 1
Clinical grounding **20**	**20 points** • Clear connection to the patient's findings, abilities and priorities • Design choices follow from the case	**15 points** • Good connection to the case • Some design choices not tied to it	**10 points** • Loose connection to the case • Could apply to any patient with this condition	**5 points** • Little or no connection to the case
Design and function **30**	**30 points** • Clear and complete design that does what it claims • Supported by a prototype, model or demonstration	**22 points** • Good design with minor gaps • Limited demonstration	**15 points** • Partial design or unclear function	**8 points** • Unclear what the device does
Usability in daily life **20**	**20 points** • Comfortable and low effort for the patient • Adapts as the condition progresses	**15 points** • Usable day to day • Adapts to some progression of the condition	**10 points** • Usable only in ideal conditions	**5 points** • Impractical for daily use
Feasibility **15**	**15 points** • Clear and realistic path to use, with cost, access and regulation considered	**11 points** • Good consideration of practical barriers with minor gaps	**8 points** • Practical barriers acknowledged but not addressed	**4 points** • Practical barriers missing or ignored
Communication **15**	**15 points** • Clear, organised and concise • Questions answered well	**11 points** • Clear overall with minor issues • Most questions answered	**8 points** • Hard to follow in places • Questions partly answered	**4 points** • Unclear or disorganised • Questions not answered

Judges score each project out of 100 with this rubric, then name the best, second best and third best project they saw in the track. The rankings from all judges determine which two teams win in each track.

PRIZES

First place

$1,000

in prizes for the top team overall

Each track has two winning teams. Track winners receive cash prizes, with amounts announced on Sunday.

External sponsor prizes

These prizes sit outside the three tracks. A team can win any of them alongside its track result. Opt in on Devpost.

Bambu Lab

Best 3D Print

Filament

IIElevenLabs

Best Use of ElevenLabs

Wireless earbuds

Gemini

Best Use of Gemini API

MLH swag kits

Entire

Best Use of Entire

Mini portable projector

SOLANA

Best Use of Solana

Unruggable hardware wallets

Tiger Data

Best Use of Tiger Data

Stream Deck Mini

VULTR

Best Use of Vultr

Assorted prizes

GoDaddy Registry

Best Domain Name from GoDaddy Registry

Digital gift card

Gift Card $100

Best vlog

$100 Amazon gift card

Vlog your weekend at BioHacks, post it on Instagram and tag @_biohacks, then drop the link in the vlogs channel on the Discord. The best vlog wins.

Workshop raffle

1 year of NordVPN, NordPass and Incogni, plus Saily eSIM data

NordVPN NordPass incogni Saily

Attend three of the four workshops and both guest lectures to enter. Six winners are drawn at the closing ceremony.

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NordVPN

NordPass

incogni

Saily

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Tiger Data

Entire

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